left child: this means that deleting `M` will be easy and recursion stops at `M`.

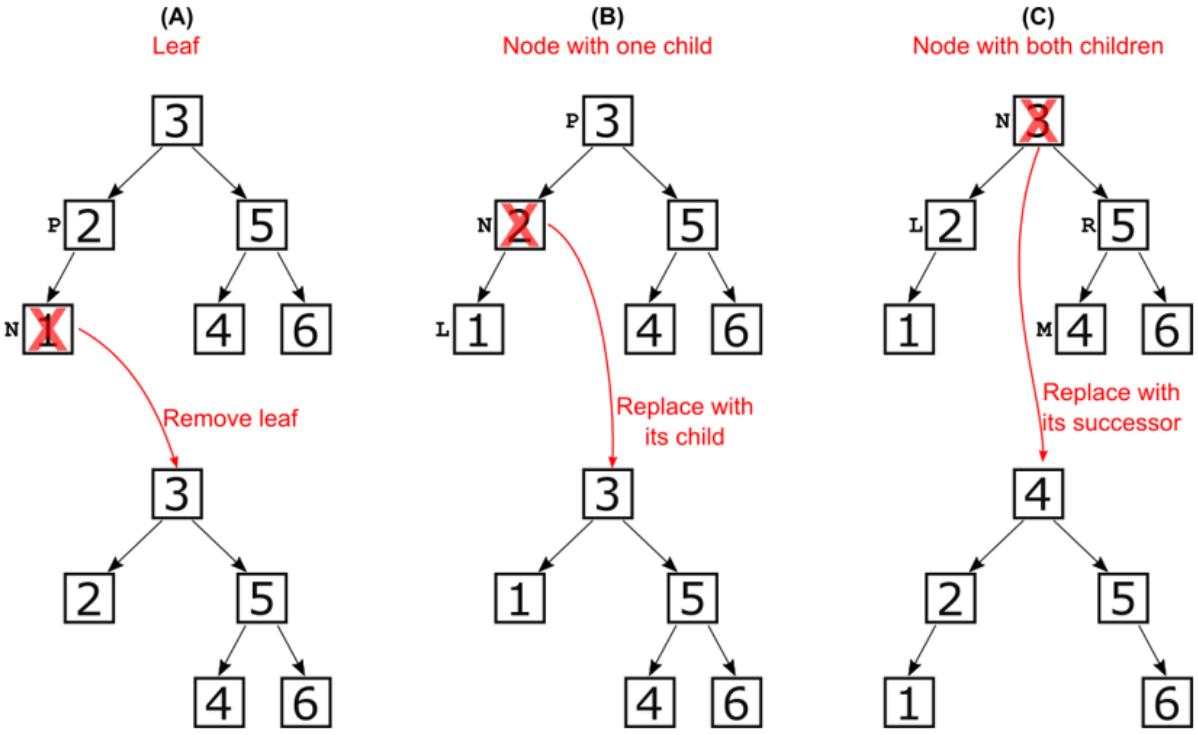


Figure 9.15 Possible cases when deleting a node from a binary search tree.
(A) Deleting a leaf.
(B) Deleting a node with a single child. (Works symmetrically if the child is in the right branch).
(C) Deleting a node with both children.

When we move from regular *BST*s to k-d trees, the first difference is caused by the fact that at each level we only use a single coordinate to partition points in the two branches. If we have to replace a node `N` at level `i`, at that level we are using coordinate `j = i mod k`, so we know its successor (for coordinate `j`) will be in `N`'s right subtree. However, when we move to `N`'s child, that node will use